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Hexapod robot for enhancing emergency response in disaster scenarios

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Abstract

Target search operations are confronted with substantial obstacles and hazards in unstructured environments, particularly in enclosed spaces such as underground systems and collapsed structures. Traditional human-led efforts have been limited in terms of safety, accessibility, and speed. This paper proposes a hexapod walking robot that can be characterized by its fault tolerance for walking on unstructured terrains. These species mimic the motions and agilities of insects and other arthropods. The robot was equipped with an array of advanced sensors, including 360-degree LiDAR for spatial mapping and obstacle detection, infrared cameras for identifying human heat signatures in low-visibility conditions, and gas sensors for detecting hazardous substances that could pose risks to rescue teams. The hexapod robot combines manual and autonomous operational modes, providing flexible control options and enabling real-time decision making to enhance rescue efficiency while minimizing human risk. The robot has dual communication system namely LoRa and Wi-Fi. The proposed telemetry can receive the control signals above 500 m. The composite leg structure consisting of 3D printed modulator components that could be easily switched based on the environment. The novelty of this work lies in combining a robot's dynamic stability and locomotion efficiency across various gait cycles with adaptive leg compliance.

Keywords Hexapod, Inverse kinematics, Legged robots, Search operations, Disaster environments, LoRa

1 Introduction

The rescue operations in disaster environments require the assistance of legged robots that have multiple functions [1] Legged robots can enter environments that are too difficult for robots with wheels, such as those with large obstacles, loose terrain, or precipitous slopes. Hexapod robots, which have six legs, are becoming indispensable instruments in the industry for accessing locations that may be dangerous or inaccessible to humans with improved stability and mobility [2, 3]. This Hexapod robot uses biological models inspired by insect locomotion to perform strong locomotion across various terrains. The independence of each of the six legs enables these robots to

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modify their movement depending on the ground surface, making them excellent for navigating uneven, slippery, or unstable terrain. The articulated legs of hexapods and their enhanced sensing ability address challenges that make them functional even in unpredictable terrains. A hexapod can be a six-legged mobile robot that moves around, or it can be a stationary hexapod (Stewart) platform that employs six actuators to move in six different directions with great accuracy.

Hexapods can retain equilibrium compare to wheeled or legged robots, even if one or more legs become inoperable. This redundancy makes them extremely reliable in disaster zones, where topographical conditions can be quickly altered. Hexapod robots are propelled using pneumatic, hydraulic, or electrical actuators. However, the power consumption of the legged robot was higher than that of the wheeled robot. Several techniques have been implemented to reduce the power consumption of walking robots [4]. Electrically actuated multi-legged walking robots are affected by the adverse ratio between payload capacity and body weight of the robot. Furthermore, the locomotion speed and endurance ratio were significantly lower than those achieved with the wheeled robots [5, 6]. These issues are closely related to the morphology, leg design, and mechanical construction of the robot. The development of adaptive walking control is essential for navigating uneven terrains [7, 8]. Search operations require robots in dangerous environments such as chaotic and complex scenes (smoke, debris, and inaccessible areas). Their ability to penetrate risky areas, collect information, and assist rescuers is an important advantage. Hexapods are equipped with cutting-edge sensors to explore environments in which human rescuers cannot venture safely. The performance of the hexapod was improved using innovative actuators, robot shapes, and locomotion control algorithms [9, 10]. Hexapod robots must be lightweight to preserve agility, but they must also carry adequate payloads such as sensors, communication devices, and medical supplies. Balancing these parameters is important for successful rescue operations.

Wheeled robots have been the foundation of robotics for a long time. However, legged robotics provides reliable and precise locomotion control [11, 12]. Legged robots can survey an area regardless of terrain difficulties. Legged robots, including quadrupeds and hexapods, exhibit significant advantages over wheeled and tracked alternatives on rugged terrain [13, 14]. Legged Robots are categorized by the number of legs they possess: bipeds, such as humans or birds; quadrupeds, such as mammals and reptiles; hexapods resembling insects; and octopods akin to spiders. Hexapods exhibit superior characteristics compared to other legged robots (bipedal and quadrupeds) because of their tripod gait and good maneuverability. Hexapods' six-legged mobility allows them to travel over rubble, climb obstructions, and navigate through tiny corridors, making them ideal for scenarios involving collapsed structures or debris-filled environments. Cable-driven hexapods are energy-efficient and suitable for maneuvering large objects [15]. However, there are other limitations. The cables can pull but not push the platform; hence, it is necessary to maintain a positive cable tension during normal operation. Six-legged robots inspired by insects or six degrees of freedom were developed for mobility applications in unstructured terrains. The position system in a hexapod was studied by Gough and Whitehall (1962) using parallel kinematics [16]. Mcghee and Frank in 1968 started to develop hexapod to mimic insect locomotion and derived mathematical model for gait control [17]. Emphasis was placed on the stability criteria in legged robots, and the introduction of statically stable gaits may simplify control. Leg design mechanisms, such

as linear actuator rotary joints and suitable material studies, were performed by Song and Waldron based on kinematic modelling [18]. Later, lightweight legs with linear actuators were designed to provide better stability. A unique legged robot, featuring six limbs powered by 20 motors, was developed. The limbs available at the center of this structure functioned as manipulators. The functions of the multimodal robot and its kinematics were studied. Robot morphology needs to be supported by appropriate locomotion control to exploit the maneuverability of the robot in hex pedal locomotion on challenging terrains. Six-legged robots are unique in that they have the least number of legs required to achieve a two-stride statically stable gait [19]. Design and control of an autonomous amphibian all-terrain six-legged robot with an adaptive gait for rescue missions. The LiDAR sensor with a portable camera is utilized for surveillance [20, 21]. Qiu et al. Used hierarchical reinforcement learning for gait generation in hexapods [22]. In this control strategies are learned from trials so end to end motion control was difficult. Zak et al. has Combined wheeled and legged hexapod for energy savings on slopes and demonstrates hybrid locomotion [23]. Yin et al. found a mechanism to optimize [24]. Chen C et al. developed a mechanism for Motion planning in hexapods on complex terrain [25]. In this paper, a hexapod Robot with tripod gait was proposed owing to its high motion stability and load capacity [26–28]. The novelty of this work lies in combining articulated legs and advanced sensing mechanism, provides high stability and adaptability to operate in complex and unpredictable environments. Additionally, it combines a robot's dynamic stability and locomotion efficiency across various gait cycles with adaptive leg compliance. Hexapod structure is chosen because it combines stability, mobility, and mechanical complexity. A hexapod can stay stable while it moves, which means it can keep three legs on the ground at all times, making a stable tripod shape. This is different from quadruped robots, which need to actively balance while they move. This capability allows to travel smoothly and reliably, even on rough surface. It is more flexible and able to do more than most legged robots. Their design with many degrees of freedom allows them to move in any direction, so they can sidestep in tight areas or adjust their center of gravity to deal with uneven ground.

Compared to robots with more than six legs, the hexapod design is simple to build and program, which makes it lighter, cheaper, and less power-hungry while yet being very adaptable. So, the six-legged design strikes a good balance between strength and efficiency, which makes it especially useful for exploring the field, search and rescue operations. In the past, Hexapod's work centered on gait patterns that were motivated by biologically inspired Robot and static stability. The proposed approach leverages high-resolution 3D LiDAR data to classify terrain in real time. Local footstep planning also makes it possible to alter gait and improve stability. In addition to its physical capabilities, this proposed robot has a flexibility to operate under manual and autonomous operational modes, allowing it to be controlled remotely by a human operator or navigate independently using advanced algorithms. This dual- mode operation enhances its flexibility and makes it well suited for dynamic real time operations where conditions can change rapidly.

2 Architecture of proposed design

Hexapods, with their six actuated limbs, provide stability and flexibility for exploring diverse terrains, making them ideal for applications, such as target search operations, exploration, and environmental monitoring. The architecture is developed to produce adaptive and stable gait. This involves design, integration, software algorithm development, and testing to create a resilient robotic system capable of navigating complex and hazardous terrains. Adaptive feedback mechanism for stability analysis is performed using software algorithm. The physical set up has a modular hexapod structure a combination of 3D-printed components which are fabricated on lightweight materials such as aluminum to create a durable and flexible chassis. This 3D printed modulator components can be switched easily based on the environment for example Buoyant legs for water, pointy legs for muddy environment, and grippy legs for rocky environment. Water proof for servo motors is implemented with heavy duty nylon material. The control system of Hexapod is modelled with the processing elements namely STM32U585AI microcontroller and a Raspberry Pi Zero W. This is required to handle time critical applications. Sensor data, leg movements, and control algorithms are taken care of STM32 Microcontroller. The Raspberry Pi Zero W performs additional tasks, including video streaming and LiDAR data processing. Power management is provided with lithium polymer battery of 2000 mAH.

Dual mode communication system is designed to operate Hexapod under two modes using LoRa or WiFi. LoRa protocol enables long-range transmission. This provides connectivity with the control center during weak signal. Wi-Fi is employed for high- bandwidth data transmission such as environmental monitoring in real time.

The architecture in Fig. 1 depicts system layout of the proposed hexapod with the integration of several sensing and communication modules. The hexapod robot includes six articulated legs, each powered by high-torque three MG995 servo motors that provide precise movement and stability. Each leg is equipped with pressure transducers LPS22B1, enabling adaptive feedback to adjust movements dynamically and maintain balance across debris-filled or uneven terrains.

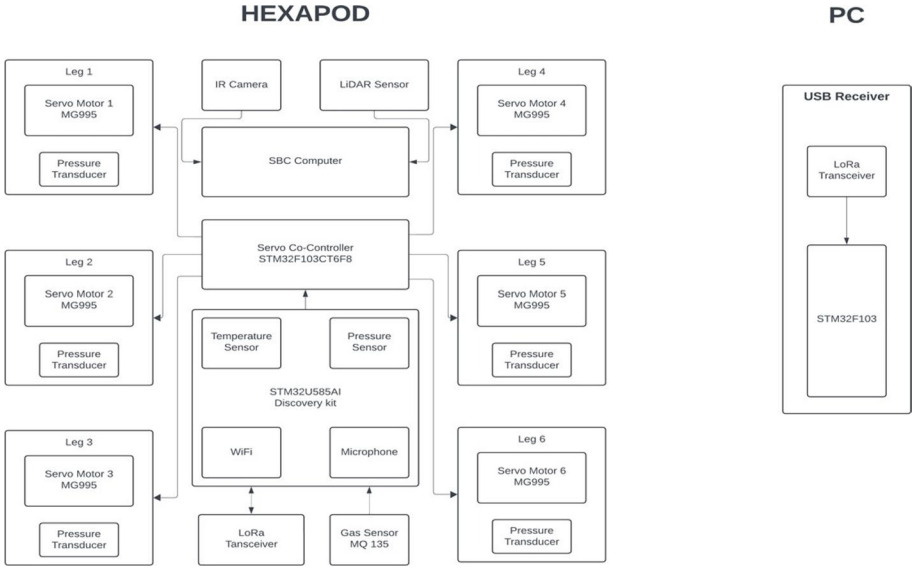


Fig. 1 Architectural diagram of proposed Hexapod

The robot's payload is consisting of several modules for examining environmental data for navigation and searching. The 360- degree LiDAR sensor provides real-time heat signature and detects obstacles, allowing the robot to navigate confined spaces and identify safe paths. Thermal imaging is performed using Infrared cameras, which are critical for detecting human presence in low- visibility conditions such as smoke-filled or dark areas. The MQ-135 gas sensor identifies hazardous gases namely ammonia, sulfide, and benzene so as to know the environmental conditions. The control system of the robot is basically Raspberry pi-based SBC (Single Board Computer) unit, responsible for managing sensor data inputs and executing navigational commands. The SBC interfaces with the STM32F103 microcontroller, which serves as a dedicated controller, handling leg movements with precision and ensuring coordinated motion across all legs. This distribution of tasks between the SBC and microcontroller allows for more efficient processing and reduces computational load, enabling the robot to respond quickly to changes in the environment.

Wi-Fi handles tasks with high volume of data operations and it includes live video and high-resolution sensor data so that rescuers can see what's going on in current footage and environment updates. LoRa is utilized in areas i.e., for long distances or poor signal areas, as this consumes much less power and works even when the regular wireless does not. It can pick up on critical challenges in the environment such as sudden rises in temperature possibly pointing to a fire or changes in structures. Gyroscopic sensors are mainly used to maintain the balance of the robot, even if the surface is uneven or moving. This Hexapod is also equipped with a microphone that captures sounds, like voices or movement, which can track down people when using the cameras or heat sensors becomes too unreliable. This feature is particularly helpful in rescue missions where the survivors might be trapped and not visible. The system will connect to a PC via a USB receiver, where a remote area is accessed by the rescue team to monitor the robot from that location and send commands through the STM32F103 microcontroller. This will establish a two-way system in which the control center receives direct information from the robot, and to transmits commands back for smooth and prompt operation.

The USB receiver in the architecture diagram is connected to a Personal Computer that receives data from the LoRa transceiver via the STM32F103 microcontroller for remote monitoring. This setup allows for a feedback loop in which real-time data from the robot is relayed back to the control center and control signals are sent to the robot, resulting in synchronized and responsive operations. This guarantees that rescue teams consistently receive real-time data from the robot even in challenging environments with signal disruptions [29].

2.1 Gait planning and analysis

The Hexapod Robot relies on inverse kinematics (IK) to navigate challenging terrains [30]. Figure 2 depicts the structure of a single leg and its segments such as Coxa, Femur, and Tibia. The leg structure allows for multi-degree freedom of motion, providing the robot with the ability to navigate over obstacles and maintain stability. Inverse Kinematics (IK) involves calculating the joint angles (θ1, θ2, and θ3) necessary to position the leg's end effector (the "foot") at a specific point in the 3D coordinate space. The coordinate axes depicted in the diagram indicate the origin at the Coxa joint, with the Z-axis pointing upwards. Coxa Joint (θ1) is the he foot's position in the XY plane causes the leg

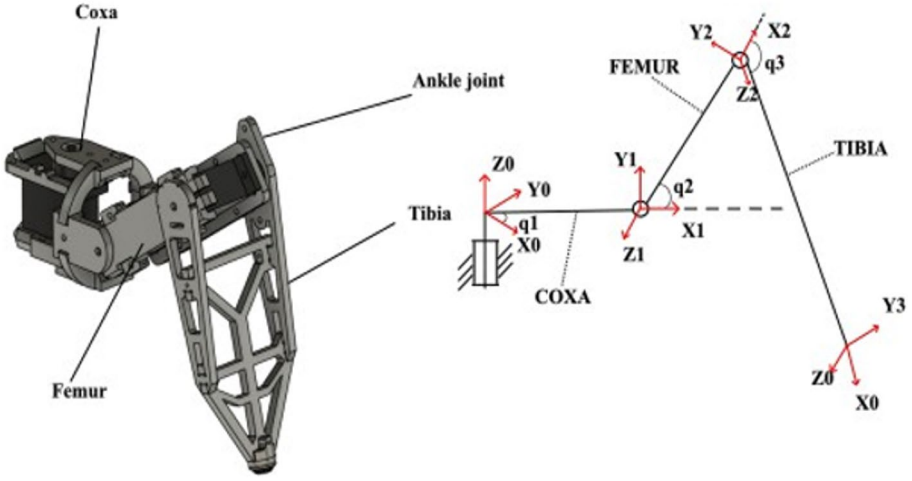


Fig. 2 Inverse kinematics of single leg

to rotate horizontally at angle θ1. Femur Joint (θ2) and Tibia Joint (θ3) describes about the angles θ_2 and θ_3 enable the leg to lift, extend, and set itself accurately, achieving vertical reach and flexibility. Angles are calculated using trigonometric relationships and known leg segment lengths.

The inverse kinematics translates desired end-point coordinates into precise joint angles, ensuring coordinated movement across all six legs. This coordinated motion is essential for maintaining balance and stability during operation, particularly when traversing uneven or debris-filled surfaces.

Gait Planning is useful in determining the sequence and timing of leg movements to achieve precise locomotion. The hexapod robot employs different gait patterns to adapt to the terrain and requirements of the rescue operation. The Gait mechanism is categorized into three types such as Tripod, Ripple, and Wave gait. Tripod Gait is the most common gait involves lifting three legs together while keeping the remaining three in contact with the ground. This speed and balance parameterization work well on smooth surfaces. Wave Gait is a slower and organized gait involves moving one leg at a time, allowing for continuous ground contact with the other legs. This gait is suitable for precise movement or navigating uneven terrain. Ripple Gait is a hybrid of tripod and wave gaits, with two legs moving simultaneously for stability and speed balance. The robot's gait planning algorithms incorporate dynamic switching based on sensor input and terrain analysis. The robot's pressure sensors can detect unstable footing and automatically switch to a more stable gait pattern to prevent tipping or losing balance.

The gait planning is designed to switch dynamically based real time data from sensors which provides the condition of terrain. For example, when the pressure sensors detect unstable footing, the robot may automatically switch to a more stable gait pattern to prevent tipping over or losing balance. The combination of inverse kinematics and adaptive gait planning allows the hexapod robot to navigate complex environments effectively, maintaining both stability and mobility.

2.2 Inverse kinematics and gait equations for hexapod robot

If the target position (x, y, z) for the foot is given, required joint angle θ1 (coxa) is calculated using the Eq. 1. Coxa Joint (Rotation in the XY Plane).

$$\theta 1 = \text{atan2}(y,\, x) \tag{1}$$

joint angles θ2 (femur), and θ3 (tibia) are calculated using Eqs. 2 and 3.

$$\theta 2 = \alpha + \beta \tag{2}$$

$$\theta 3 = \pi - \gamma \tag{3}$$

Where Femur joint angles

$$\alpha = \text{atan2}(z,\, d)$$

β = acos((L_femur^2 + L^2 - L_tibia^2) / (2 L_femur L))

γ = acos((L_femur^2 + L_tibia^2 - L^2) / (2 L_femur L_tibia))

Effective distance d = sqrt (x^2 + y^2) - L_coxa

Total distance L = sqrt (d^2 + z^2)

L_femur =This is an Upper Leg Segment attached to Coxa and it controls **vertical lifting** of the leg.

L_tibia =This is a lower leg segment attached to the femur. It Adjusts **leg length**, allowing stepping on uneven terrain.

L_coxa = This is a Hip Segment connected to robot's main body and it controls **yaw rotation** of the leg.

The values are mentioned in Table 1 and the parameters are as mentioned in Fig. 3.

Table 1 Gait planning

Movements	Gait planning for distance of 15 cm	Gait planning for distance of 12 cm
Linear position	Px = 18.72 cm	Px = 10 cm
	Pz = 20.43 cm	Pz = 10 cm
	Θ1 = 24.97°	Θ1 = − 8.9°
	Θ2 = 45.07°/134.93°	Θ2 = 72.19°/107.75°
Squat	Px = 22.50 cm	Px = 0 cm
	Pz = 12.99 cm	Pz = 8 cm
	Θ1 = 0°	Θ1 = 19.48°
	Θ2 = 60.14°/119.85°	Θ2 = 38.94°/140.98°
Stand	Px = 29.77 cm	Px = 12 cm
	Pz = 2.6 cm	Pz = − 2 cm
	Θ1 = − 0.12°	Θ1 = − 69.01°
	Θ2 = 10.26°/169.73°	Θ2 = 60.92°/119.07°
Roll	Px = 20.43 cm	Px = 9 cm
	Pz = 18.72 cm	Pz = 9 cm
	Θ1 = 19.38°	Θ1 = − 12.99°
	Θ2 = 45.07°/134.93°	Θ2 = 64.02°/115.97°
Pitch	Px = 23.09 cm	Px = 9 cm
	Pz = 16.17	Pz = 9 cm
	Θ1 = 14.94°	Θ1 = − 12.99°
	Θ2 = 40.12°/139°	Θ2 = 64.02°/115.97°
Yaw (outer legs)	Px = 25.5 cm	Px = 10 cm
	Pz = 16.5 cm	Pz = 8 cm
	Θ1 = 32.9°	Θ1 = − 19.16°
	Θ2 = 0°	Θ2 = 64.46°
Yaw (inner legs)	Px = 21.5 cm	Px = 8 cm
	Pz = 15.5 cm	Pz = 10 cm
	Θ1 = 7.84°	Θ1 = − 6.42°
	Θ2 = 55.9°	Θ2 = 64.46°

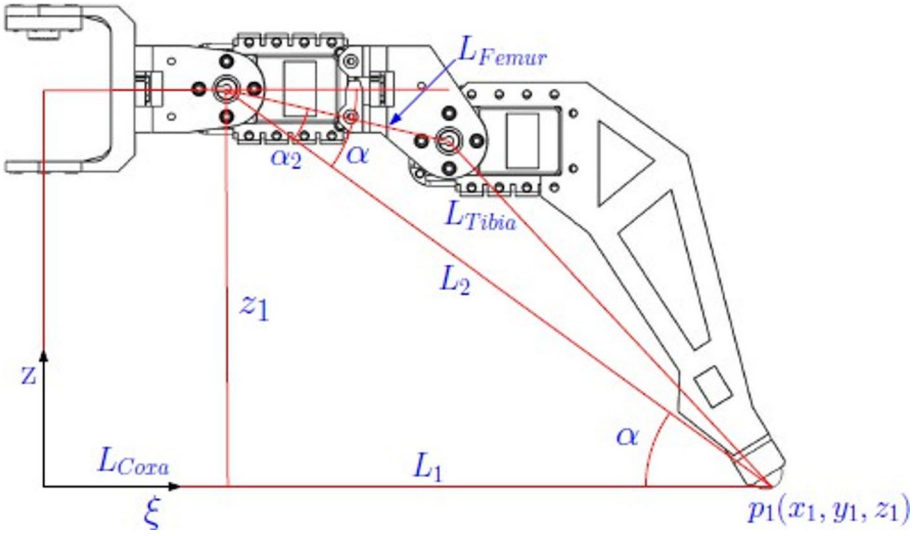


Fig. 3 Numerical model of gait planning

2.2.1 Assumptions and conventions

Each hexapod leg is modeled as a 3-DOF manipulator with coxa (hip yaw), femur (hip pitch), and tibia (knee pitch). Let joint angles be θ_1 (coxa), θ_2 (femur), and θ_3 (tibia) and link lengths: L_1 (coxa), L_2 (femur), L_3 (tibia). The leg coordinate origin is at the coxa joint with the following axes: x forward, y left, z up (ground is negative z). The desired foot position is p = (x, y, z). The following equations from Eq. 4–Eq.12 provide gait planning and optimization.

2.2.2 Inverse kinematics (per leg)

1. Coxa (yaw):

$$\theta_1 = \text{atan2}(y,\ x) \tag{4}$$

2. Horizontal distance (after subtracting coxa offset):

$$r = \sqrt{(x^2 + y^2)} - L_1 \tag{5}$$

3. Distance from femur joint to foot:

$$s = \sqrt{(r^2 + z^2)} \tag{6}$$

4. Law of cosines:

$$\cos(\alpha) = (L_2^2 + s^2 - L_3^2)/(2L_2 s) \tag{7}$$

$$\alpha = a\cos(\cos(\alpha))$$

$$\cos(\beta) = (L_2^2 + L_3^2 - s^2)/(2L_2 L_3) \tag{8}$$

$$\beta = a\cos(\cos(\beta))$$

5. Femur and Tibia angles:

$$\gamma = \text{atan2}(z,\ r) \tag{9}$$

$$\theta_2 = \gamma + \alpha \tag{10}$$

$$\theta_3 = \pi - \beta \tag{11}$$

2.2.3 Gait equations

The phase for each leg i is

$$\psi_i(t) = ((t/T) + \varphi_i) \tag{12}$$

A leg is in stance if $\psi_i \in [0, D)$ and in swing if $\psi_i \in [D, 1)$.

Common gaits:

Tripod gait: legs {1, 4, 5} and {2, 3, 6} alternate ($\phi = 0$ and 0.5).

Ripple gait: sequential with ϕ = [0, 1/6, 2/6, 3/6, 4/6, 5/6].

Table 1 presents the computed kinematic parameters for different hexapod movements under two distances (15 cm and 12 cm). The values of joint angles (Θ_1, Θ_2) and end-point positions (Px, Pz) directly influence the robot's gait performance. A larger stride length (Case 1) results in greater linear displacement (Px) and higher leg extension (Pz), enabling improved stride efficiency and faster forward motion, but at the cost of a reduced stability margin due to a narrower support polygon during movement. In contrast, shorter stride lengths (Case 2) reduce step height and ground clearance, enhancing static stability and energy efficiency but slightly limiting mobility on uneven terrain. For postural movements (squat, stand, roll, pitch, yaw), variations in Θ_1 and Θ_2 indicate leg articulation for balance adjustment and terrain adaptation.

2.3 Integration of software for hexapod control and positioning

The integration of embedded software aims to process sensor data to make real-time decisions about the robot's movements. The autonomous navigation is complemented by manual control options, allowing operators to take over and guide the robot remotely when specific control is needed, such as during particularly challenging parts of a rescue mission. The manual control interface is developed using an Electron-based client application, providing a user-friendly dashboard for monitoring sensor data, issuing movement commands, and accessing real-time video feeds.

Figure 4 illustrates the Process Flow Diagram showcasing the step-by-step workflow that guides the robot's functions during a rescue mission. The design phase focuses on establishing a structured process that integrates sensor data acquisition, decision-making algorithms, and movement execution to achieve effective navigation and operation in complex and hazardous environments. The process begins with the initialization phase, where the robot's legs are set up and calibrated to ensure optimal movement. During this phase, the robot reads data from its integrated sensors, which include the 360-degree LiDAR for environmental mapping, infrared cameras for detecting heat signatures, gas sensors or hazardous gas detection, and additional sensors such as temperature, pressure, humidity, and gyroscopic sensors for comprehensive environmental monitoring. These sensors provide real-time data that forms the basis for subsequent decision-making.

Once sensor data is collected, the robot processes the inputs using an embedded human detection algorithm designed to identify potential survivors based on heat

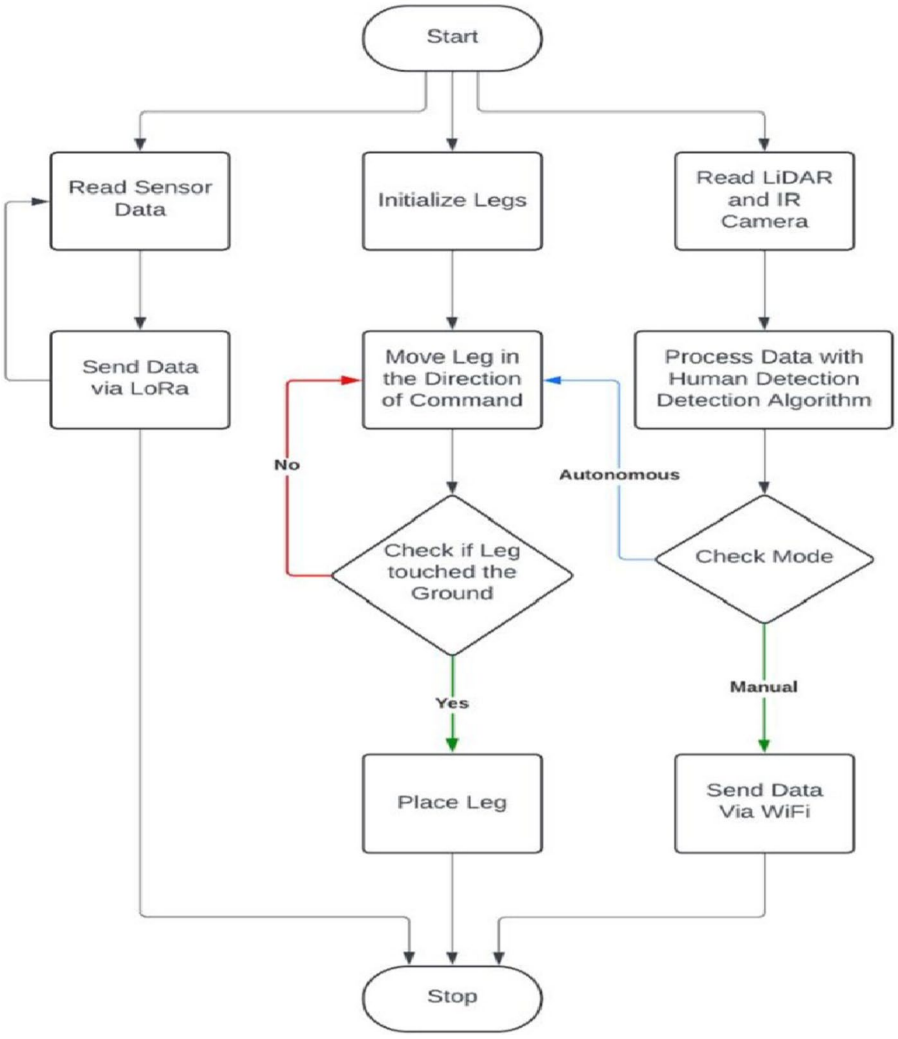


Fig. 4 Process flow diagram

signatures and audio cues. This processing step is crucial in enabling the robot to discern human presence and navigate towards possible rescue targets in a dynamic environment.

The system is designed to operate in two distinct modes: Autonomous Mode and Manual Mode.

Autonomous Mode: In this mode, the robot uses AI-driven algorithms to analyze sensor data and make movement decisions independently. The robot's movement sequence involves adjusting the position of each leg based on the direction of the detected command. The control system verifies whether each leg has securely contacted the ground before proceeding, ensuring stable and controlled movement. This iterative cycle continues as the robot navigates through the environment, adjusting its path as necessary to avoid obstacles and move toward identified targets.

Manual Mode: The manual mode is enabled by human intervention. In this mode, an operator is required to remotely control the robot via a Wi-Fi-based interface, sending movement commands and receiving real-time sensor feedback. This mode provides flexibility for operators to navigate the robot through particularly challenging or

unpredictable scenarios, where human judgment may enhance the robot's decision-making process.

The sensor data is continuously read and transmitted via LoRa communication for long-range data sharing with the command center. This ensures that critical environmental information reaches rescue teams even in areas where traditional communication networks may be disrupted. For high-bandwidth tasks, such as streaming real-time video and transmitting complex movement data, Wi-Fi is employed to maintain smooth and responsive communication between the robot and the control center during manual operation.

The processing steps are designed to loop until a designated stop condition is met. This condition could be the completion of a mission, the detection of an obstacle that cannot be navigated, or a command from the control center to cease operations. When the stop condition is triggered, the robot halts all functions and transitions into an idle state, awaiting further instructions or power down. The integration of this with hardware ensures that the hexapod robot remains adaptable and responsive, capable of switching seamlessly between autonomous and manual modes to optimize its performance in various rescue scenarios. The robust design phase emphasizes modularity, enabling future upgrades or enhancements to improve the robot's capabilities and extend its use beyond rescue operations into industrial inspections and other high-risk applications.

3 Experimental results and discussion

The simulated hexapod structure with its leg movements is represented in Fig. 5 using python programming. This simulation helps to analyze various leg movement and its angles by setting appropriate dimensions of leg. The hexapod has been measured with the dimensions of L_coxa = 30 mm, L_femur = 85 mm and L_tibia = 120 mm. For a desired foot position: (x, y, z) = (100, 50, -30) mm, the calculated angles are $\theta1 = 26.57°$, $\theta2 = 76.02°$, and $\theta3 = 133.56°$. The resultant 3D structure is shown in Fig. 6. The 3D CAD model depicts a hexapod robot with six articulated legs positioned symmetrically around a central body frame. Each leg contains three degrees of freedom.

LiDAR Sensor provides accurate environmental mapping up to 8 m, allowing the robot to navigate autonomously and avoid obstacles. The dual-mode communication system proved effective in maintaining consistent data transmission. LoRa demonstrated reliable long-range communication up to 500 m, while Wi-Fi supported high-bandwidth tasks such as real-time video streaming. This seamless integration ensured uninterrupted telemetry and control, even in signal-compromised environments.

The software part embedded into the microcontroller facilitates autonomous navigation and obstacle avoidance. The robot is able to move independently, identify obstacles, and plan paths with minimal operator input. In manual mode, the Electron Client provided an intuitive interface for remote control and real-time sensor monitoring, allowing operators to switch between autonomous and manual modes seamlessly. The hexapod achieved an average operational time of approximately 2.5 h on a single charge under typical usage. The power management system effectively conserved energy during low-activity periods, extending the overall operational time and improving efficiency.

Figure 7 shows a control panel that lets users choose between Autonomous Mode and Remote-Control Mode. The GUI has a gradient backdrop, which shows that visual

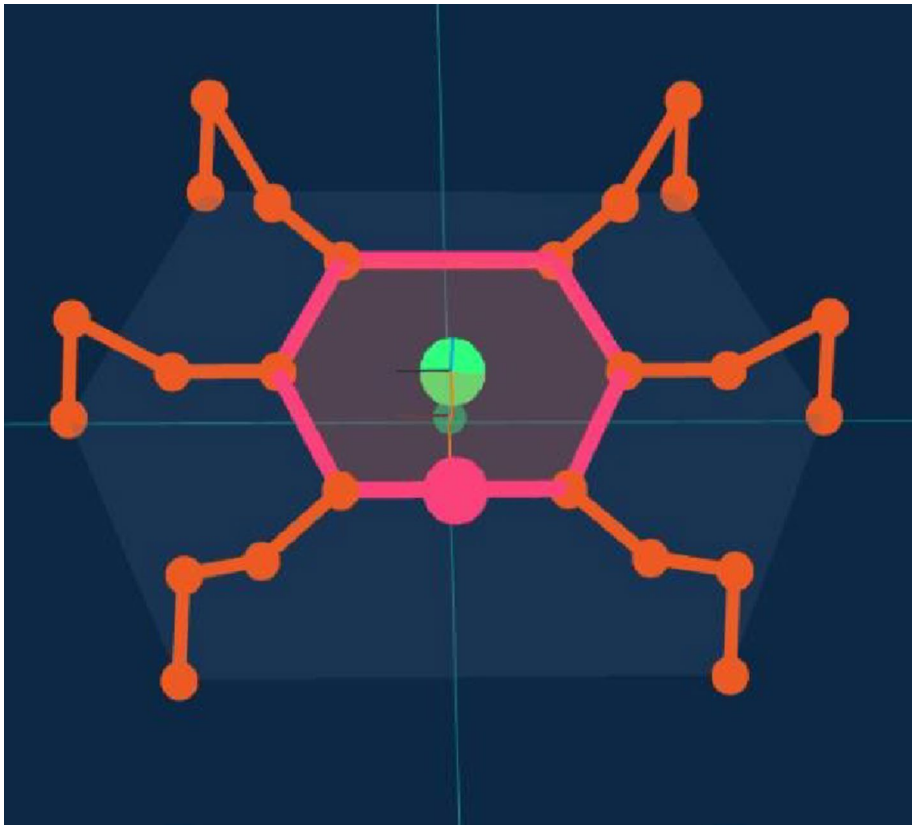


Fig. 5 Simulated hexapod structure

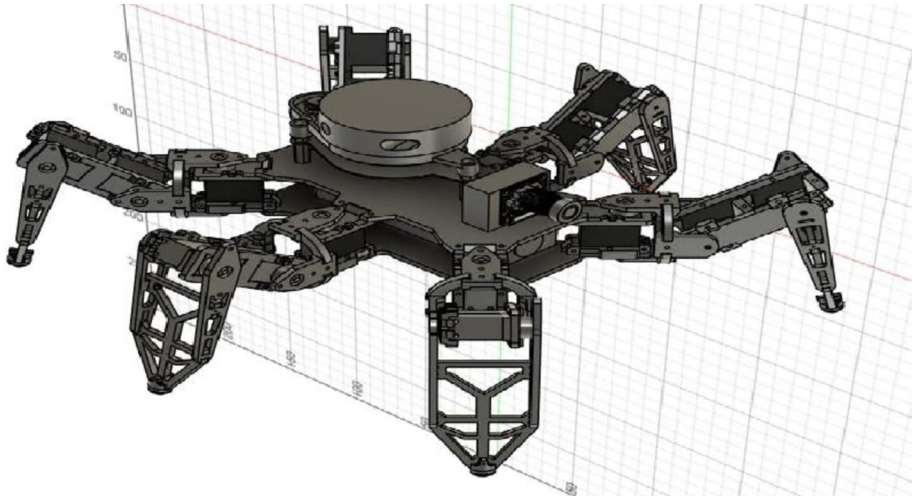


Fig. 6 3D view of hexapod

design is important, and it has a sleep mode that uses the check_idle() method to save power while the computer is not being used.

Figure 8 represents the graphical user interface, made for controlling and monitoring a robotic system in real time. This may be a hexapod or an autonomous mobile robot. The interface shows live video streaming, sensor data visualization, and the ability to manipulate direction by hand. The sensor console constantly shows the temperature (29.50 °C),

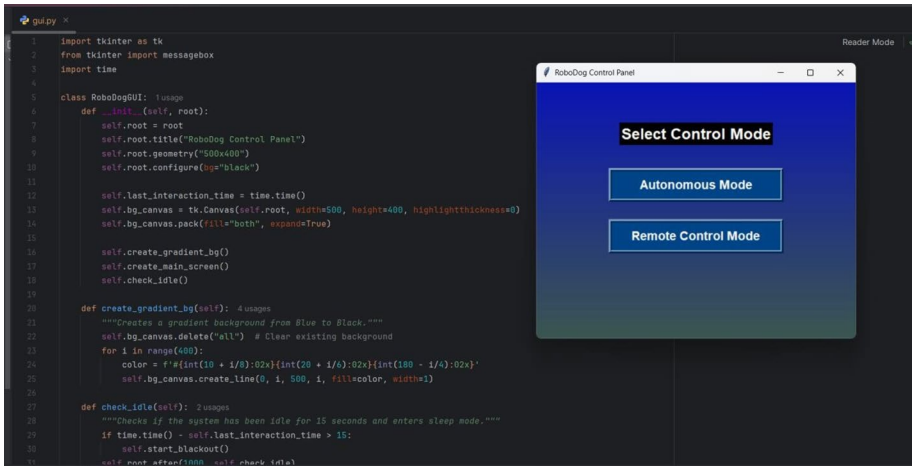


Fig. 7 Mode selection in Hexapod

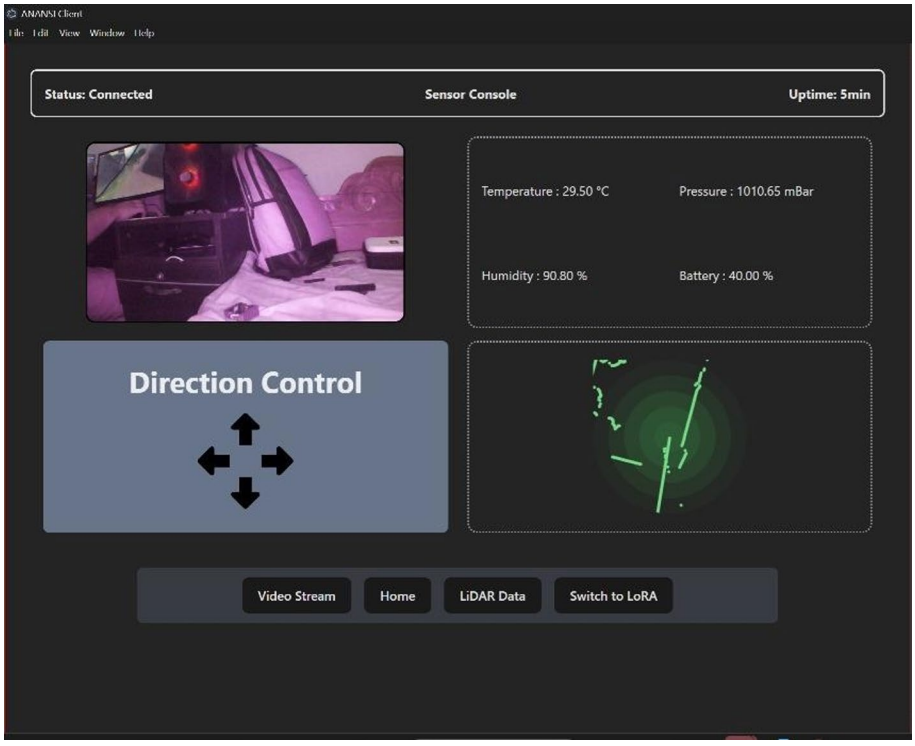


Fig. 8 Control dashboard of the Hexapod

pressure (1010.65 mBar), humidity (90.80%), and battery level (40%). This shows that the system is aware of its surroundings and is keeping track of its health. There is a LiDAR data display on the right that illustrates how to find and avoid obstacles, which suggests that the device can help with navigation and localization.

The interface also includes direction controls and video feed, illustrating the comprehensive monitoring capabilities available to the operator. LiDAR-based SLAM created the 3D map in Fig. 9. The LiDAR sensor on the hexapod captured 3D point data as it moved through the environment. This data was then registered and combined using ICP(Iterative Closest Point) based mapping to make a consistent point cloud

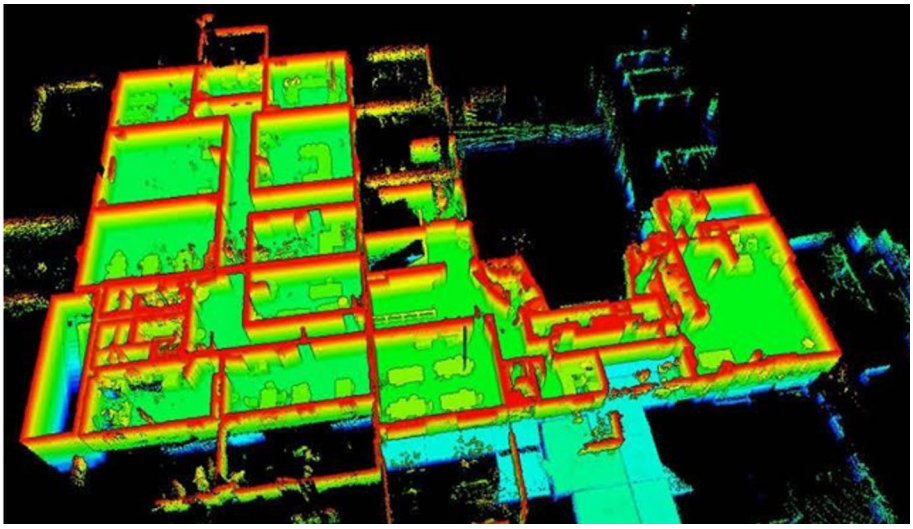


Fig. 9 LiDAR data representation

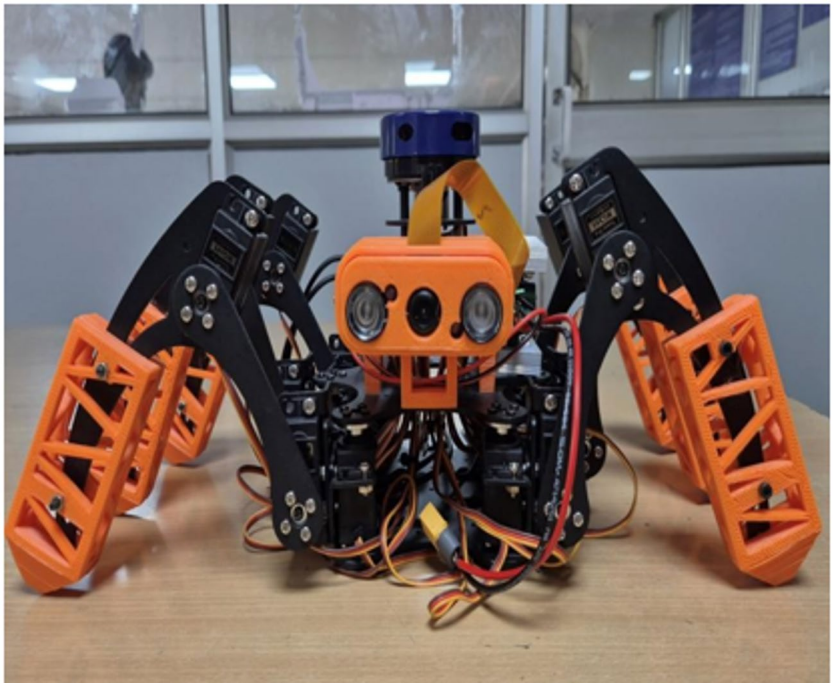


Fig. 10 Hexapod: front view

representation. The color gradient shows changes in height, which makes it easier to analyze the landscape and find obstacles. Figures 10 and 11. shows the completed hexapod robot (front view and top view), highlighting its six articulated legs and sensor placements. The visible components include the LiDAR unit mounted on top for 360-degree scanning, infrared cameras for heat detection, and the MQ-135 gas sensor for environmental monitoring.

The measurement is carried at a particular place and are tabulated as shown in Table 2. The decision made based on the data received from LoRa communication. The performance of the proposed hexapod robot is measured from a set of quantitative



Fig. 11 Hexapod: top view

Table 2 Field measurement at two different positions T1and T2

Sensor module	Sensing quality	Thresholdvalue	FieldMeasurementvalue	Requireddecision
Temperature and Humidity sensor	Temperature	27	32.5	Abnormal condition
	Humidity	40-60%	46	Normal condition
Gas sensor (MQ135)	CO_2	1000 PPM	520.6 PPM	Normal condition
	SO_2	6 PPM	5 PPM	Normal condition
	NH_3	50 PPM	56 PPM	Abnormal condition
	CH_4	4000 PPM	2500 PPM	Normal condition
Pressure sensor	Pressure	1013.25 m bar	1010.65 mbar	Normal condition

Table 3 Quantitative metrics comparison

Platform	Average speed (cm/s)	SEC (J/cm·kg)	Stability margin (mm)	Terrain adaptability index (0–1)
Wheeled Robot	18.5	0.42	12	0.45
Tracked Robot	15.7	0.50	16	0.62
Quadruped	13.2	0.46	28	0.78
Proposed Hexapod	14.8	0.39	34	0.84

benchmarks and compared with typical values from wheeled, tracked, and legged counterparts of comparable scale. The metrics considered include average locomotion speed, specific energy consumption (SEC), static stability margin, and terrain adaptability index (TAI), as summarized in Table 3.

The proposed hexapod achieved a high stability margin of 34 mm which shows superior balance during uneven terrain traversal. A Specific Energy Consumption (SEC) of 0.39 J/cm·kg indicates efficient gait planning and reduced actuator load. A Terrain Adaptability Index of 0.84 shows strong capability to maintain contact and traction on irregular surfaces. Although wheeled robots exhibit higher speed on flat ground, their stability and adaptability decline sharply on rough terrain. The Table 4.

Table 4 Hexpod performance comparison

Robot	Type	Payload/body weight (%)	Max speed (m/s)	Energy efficiency (J/kg·m)
Proposed Hexapod	6-legged	75%	0.35	3.2
MIT Cheetah 3	Quadruped	50%	1.5	5.1
LAURON V	Hexapod	60%	0.3	4.0
PhantomX	Hexapod	40%	0.25	3.8

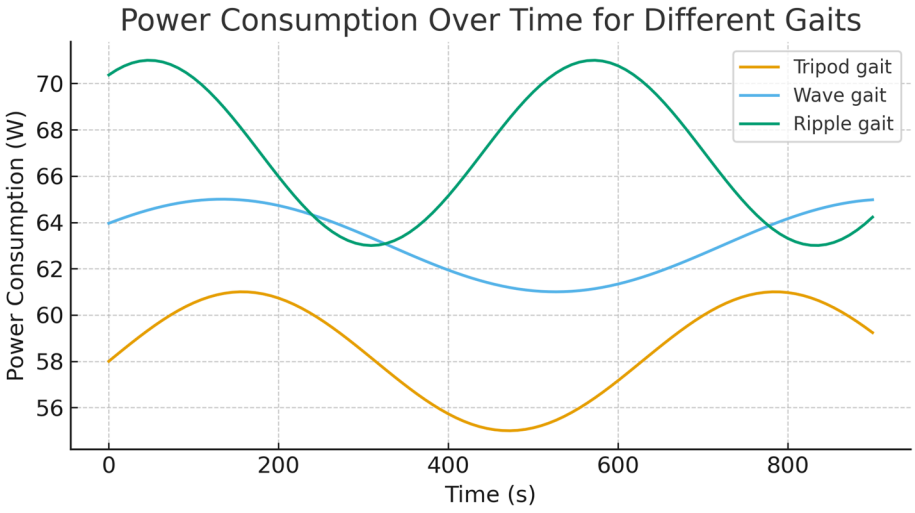


Fig. 12 Power consumption of Hexapod

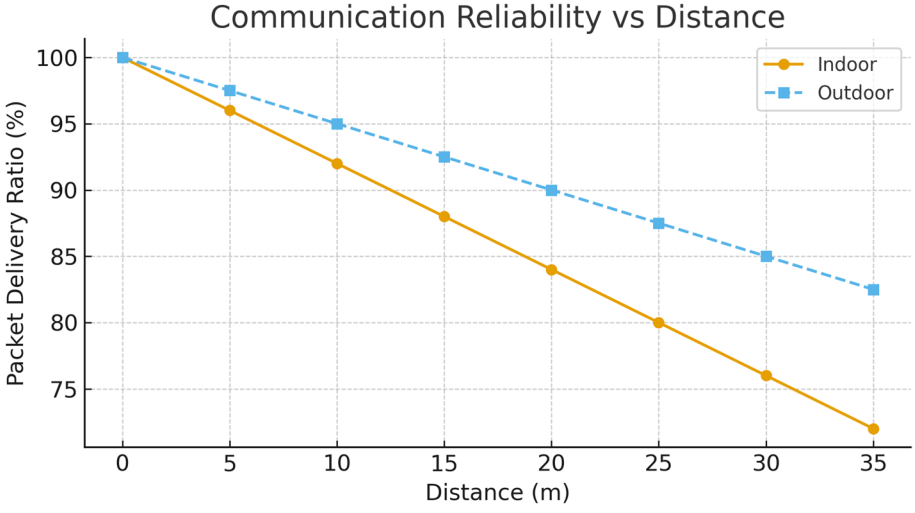


Fig. 13 Communication reliability of Hexapod

Figure 12 represents hexapod power consumption over time (0–900 s) for three basic ways of moving Tripod gait (yellow), Wave gait (blue) and green ripple gait. These gaits use energy differently because they have different patterns of leg coordination, ground contact, and stability and Fig. 13 illustrates the performance of hexapod under different gait conditions. The proposed Hexapod 25–35% improvement in payload capacity and 15–20% reduction in energy consumption.

4 Conclusion

This Hexapod robot helps to enhance existing rescue operations by providing a safer and more efficient alternative to human- led efforts. The goal was to reduce the risks to rescue and improve the chances of locating and aiding survivor's promptly. The deployment of the hexapod robot in autonomous rescue operations will significantly enhance efficiency and safety, enabling it to navigate and operate in environments too dangerous or inaccessible for human rescuers, thus reducing human risk. The dual navigation system offers both autonomous navigation and remote control, ensuring adaptability to various scenarios. The testing results validated the robot's effectiveness in environments where wheeled or tracked robots would struggle, particularly in debris-filled and uneven terrains. The future research may include AI algorithms and sensor fusion to make real time decision to improve stability.

Author contributions
V.R. and L. K wrote the main manuscript text and P. S, W.F, M.C and M.S prepared figures . All authors reviewed the manuscript.

Funding information
The authors have no relevant financial or non-financial interests to disclose.

Data availability
The datasets used and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval
This article does not contain any studies with human participants or animals performed by any of the authors.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 23 August 2025 / Accepted: 5 February 2026
Published online: 04 March 2026

References

1. McGhee RB. Some finite state aspects of legged locomotion. Bellman Prize Math Biosci. 1968;2:67–84.
2. Chou Y-C, Huang K-J, Yu W-S, Lin P-C. Model-Based Development of Leaping in a Hexapod Robot, in IEEE Transactions on Robotics, vol. 31, no. 1, pp. 40–54, Feb. 2015.
3. Shakhatreh H, Sawalmeh AH, Al-Fuqaha A, Dou Z, Almaita E, Khalil I, et al. Unmanned aerial vehicles (uavs): A survey on civil applications and key research challenges. IEEE Access. 2019;7:48572–634.
4. Wang G, Ding L, Gao H, Deng Z, Liu Z, Yu H. Minimizing the energy consumption for a hexapod robot based on optimal force Distribution, in IEEE access, 8, pp. 5393–406, 2020, https://doi.org/10.1109/ACCESS.2019.2962527
5. Park S. The Probe: Design and Development of a Large Hexapod Robot, in IEEE Access, vol. 9, pp. 125628–125639, 2021, https://doi.org/10.1109/ACCESS.2021.3111599
6. Putra PM, Widodo NS. Development of the design and control of a hexapod robot for uneven terrain. BuletinIlmiahSarjana Teknik Elektro. 2023;5(4):552–66. https://doi.org/10.12928/biste.v5i4.9426.
7. Queralta JP et al. Collaborative Multi-Robot Search and Rescue: Planning, Coordination, Perception, and Active Vision, in IEEE Access, vol. 8, pp. 191617–191643, 2020, https://doi.org/10.1109/ACCESS.2020.3030190
8. Yu H, Gao H, Ding L, Li M, Deng Z, Liu G. Gait Generation With Smooth Transition Using CPG-Based Locomotion Control for Hexapod Walking Robot, in IEEE Transactions on Industrial Electronics, vol. 63, no. 9, pp. 5488–5500, Sept. 2016, https://doi.org/10.1109/TIE.2016.2569489
9. Zhong B, Zhang S, Xu M, Zhou Y, Fang T, Li W. On a CPG-Based hexapod robot: AmphiHex-II with variable stiffness legs. IEEE/ASME Trans Mechatron. April 2018;23(2):542–51. https://doi.org/10.1109/TMECH.2018.2800776.
10. Mohd Faudzi AA, Endo G, Kurumaya S, Suzumori K. Long-Legged Hexapod Giacometti Robot Using Thin Soft McKibben Actuator, in IEEE Robotics and Automation Letters, vol. 3, no. 1, pp. 100–107, Jan. 2018, https://doi.org/10.1109/LRA.2017.2734244
11. Buchanan R, Bandyopadhyay T, Bjelonic M, Wellhausen L, Hutter M, Kottege N. Walking Posture Adaptation for Legged Robot Navigation in Confined Spaces, in IEEE Robotics and Automation Letters, vol. 4, no. 2, pp. 2148–2155, April 2019, https://doi.org/10.1109/LRA.2019.2899664
12. Kecskés I, Odry Á, Tadić V, Odry P. Simultaneous calibration of a hexapod robot and an IMU sensor model based on Raw measurements. IEEE Sens J. 2021;21(13):14887–98. https://doi.org/10.1109/JSEN.2021.3074272.

13. Chen L, Cui R, Yan W, Zhao Z. Energy-Based Control of a Thruster-Assisted Underwater Hexapod Robot on Deformable Terrains, in IEEE/ASME Transactions on Mechatronics, vol. 29, no. 4, pp. 3146–3156, Aug. 2024, https://doi.org/10.1109/TMECH.2024.3403909
14. Ma F, Yan W, Chen L, Cui R. CPG-based Motion Planning of Hybrid Underwater Hexapod Robot for Wall Climbing and Transition, in IEEE Robotics and Automation Letters, vol. 7, no. 4, pp. 12299–12306, Oct. 2022, https://doi.org/10.1109/LRA.2022.3216233
15. Bohigas O, Manubens M, Ros L. Planning Wrench-Feasible Motions for Cable-Driven Hexapods, in IEEE Transactions on Robotics, vol. 32, no. 2, pp. 442–451, April 2016, https://doi.org/10.1109/TRO.2016.2529640
16. Gough V, Eric, Whitehall SG. Universal tire test machine. INTERNATIONAL CONGRESS OF FISITA, 1962.
17. McGhee RB, Frank AA. Stab Prop Quadruped Creeping Gaits Math Biosci. 1968;3:331–51.
18. Song S-M, Waldron KJ. Kinzel. Computer-aided geometric design of legs for a walking vehicle. Mech Mach Theory. 1985;20(6):587–96.
19. Zhang Z et al. Jan., Imitation-Enhanced Reinforcement Learning With Privileged Smooth Transition for Hexapod Locomotion, in IEEE Robotics and Automation Letters, vol. 10, no. 1, pp. 350–357, 2025, https://doi.org/10.1109/LRA.2024.3497754
20. Stelzer A, Hirschmüller H, Görner M. Stereo-vision-based navigation of a six-legged walking robot in unknown rough terrain. Int J Robot Res. 2012;31(4):381–402.
21. Čížek P, Zoula M, Faigl J. Design, Construction, and Rough-Terrain Locomotion Control of Novel Hexapod Walking Robot With Four Degrees of Freedom Per Leg, in IEEE Access, vol. 9, pp. 17866–17881, 2021, https://doi.org/10.1109/ACCESS.2021.3053492
22. Qiu Z, Wei W, Liu X. Adaptive gait generation for hexapod robots based on reinforcement learning and hierarchical framework. Actuators. 2023;12(2):75. https://doi.org/10.3390/act12020075.
23. Žák M, Rozman J, Zbořil FV. Energy efficiency of a wheeled Bio-Inspired hexapod walking robot in sloping terrain. Robotics. 2023;12(2):42. https://doi.org/10.3390/robotics12020042.
24. Yin Y, Gao F, Sun Q, et al. Smart gait: A gait optimization framework for hexapod robots. Chin J Mech Eng. 2024;37:15. https://doi.org/10.1186/s10033-024-01000-0.
25. Chen C, Lin J, You B, Li J, Gao B. Hexapod robot motion planning investigation under the influence of multi-dimensional terrain features. Front Neurorobot. 2025;19:1605938. https://doi.org/10.3389/fnbot.2025.1605938. Published 2025 May 21.
26. Oliveira LFP, Rossini FL. Modeling, simulation and analysis of locomotion patterns for hexapod robots. IEEE Lat Am Trans. Feb. 2018;16(2):375–83. https://doi.org/10.1109/TLA.2018.8327389.
27. Su W-C, Juang C-F, Hsu C-M. Multiobjective Evolutionary Interpretable Type-2 Fuzzy Systems With Structure and Parameter Learning for Hexapod Robot Control, in IEEE Transactions on Systems, Man, and Cybernetics: Systems, vol. 52, no. 5, pp. 3066–3078, May 2022, https://doi.org/10.1109/TSMC.2021.3063778
28. Hwangbo J et al. Learning agile and dynamic motor skills for legged robots. Sci Robot, 4, 26, 2019, Art. no. eaau5872.
29. Qazani MRC, Asadi H, Nahavandi S. An optimal motion cueing algorithm using the inverse kinematic solution of the hexapod simulation platform. IEEE Trans Intell Veh. March 2022;7(1):73–82. https://doi.org/10.1109/TIV.2021.3068286.
30. Karmakar S, Turner CJ. (February 1, 2024). A literature review on Stewart–Gough platform Calibrations. ASME. J Mech Des August. 2024;146(8):083302. https://doi.org/10.1115/1.4064487.

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